Modelling Fluid-Structure Interaction using Smoothed Particle Hydrodynamics

SHTC aproach

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What is the Thesis about?

Can we describe fluid-structure interaction problems using a single system of equations for both the fluid and the solid within a single numerical framework?

SHTC

SHTC - GPR model - Godunov Peskhov Romenski

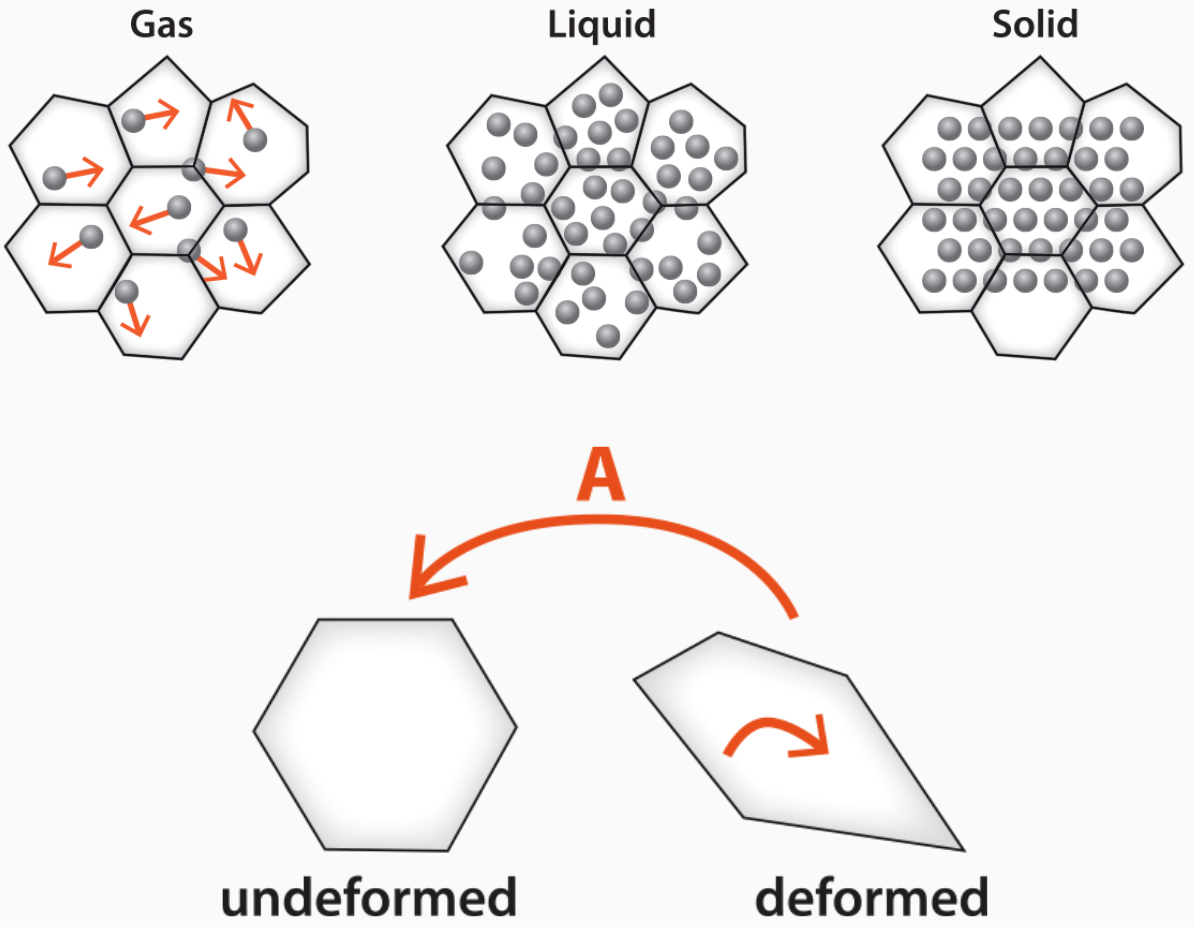
It introduces the matrix distortion field **A** and relaxation time τ to describe the material.

Loosely speaking the *relaxation time* τ describes how long bonds between neighbouring particles last i.e. how long are the particles in contact with each other.

- $\tau = 0$ ideal fluids
- $0 < \tau < \infty$ viscous fluids
- $\tau = \infty$ elastic solids

The fluidity is introduced via relaxation of distortion matrix **A**.

SHTC



SHTC - GPR model

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{1}$$

Momentum equation:

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\mathbf{v} \otimes \rho \mathbf{v} - \mathbf{S}) = 0 \tag{2}$$

Equation for distortion:

$$\frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{A} + \mathbf{A} \nabla \mathbf{v} = -\frac{1}{\theta} E_{\mathbf{A}} \tag{3}$$

Equation for energy:

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot (\mathbf{v} \rho E - \mathbf{S} \mathbf{v}) = 0 \tag{4}$$

$\mathbf{A}$ is the distortion field, $\mathbf{S}$ is the stress tensor, $\theta = \theta(\rho, \tau, \mathbf{A}) \geq 0$ is a *relaxation function*, $E = E(\rho, \mathbf{v}, \mathbf{A}) = \epsilon(\rho, \mathbf{A}) + \frac{1}{2}\|\mathbf{v}\|^2$ is total energy density.

SPH

About SPH - Smoothing Kernels

What is a smoothing kernel? It is an approximation of **Dirac delta function** δ. More specifically Smoothing Kernel $W_h(\mathbf{x})$ is function: $\mathbf{R}^d \to [0, \infty)$ with the following properties:

- $\int W_h(\mathbf{x}) d\mathbf{x} = 1$
- W_h is sufficently smooth ($\mathbf{C}^2$ is usually enough)
- W_h is radial function i.e. $W_h(\mathbf{x}) = W_h(|\mathbf{x}|)$
- $W_h(r) = \frac{1}{h^d} W_1(\frac{r}{h})$
- $\frac{dW_h}{dr} \leq 0$

Parameter h is called *smoothing length* and it holds that $W_h(r) \to \delta(r)$ as $h \to 0^+$.

About SPH - Smoothing Kernels

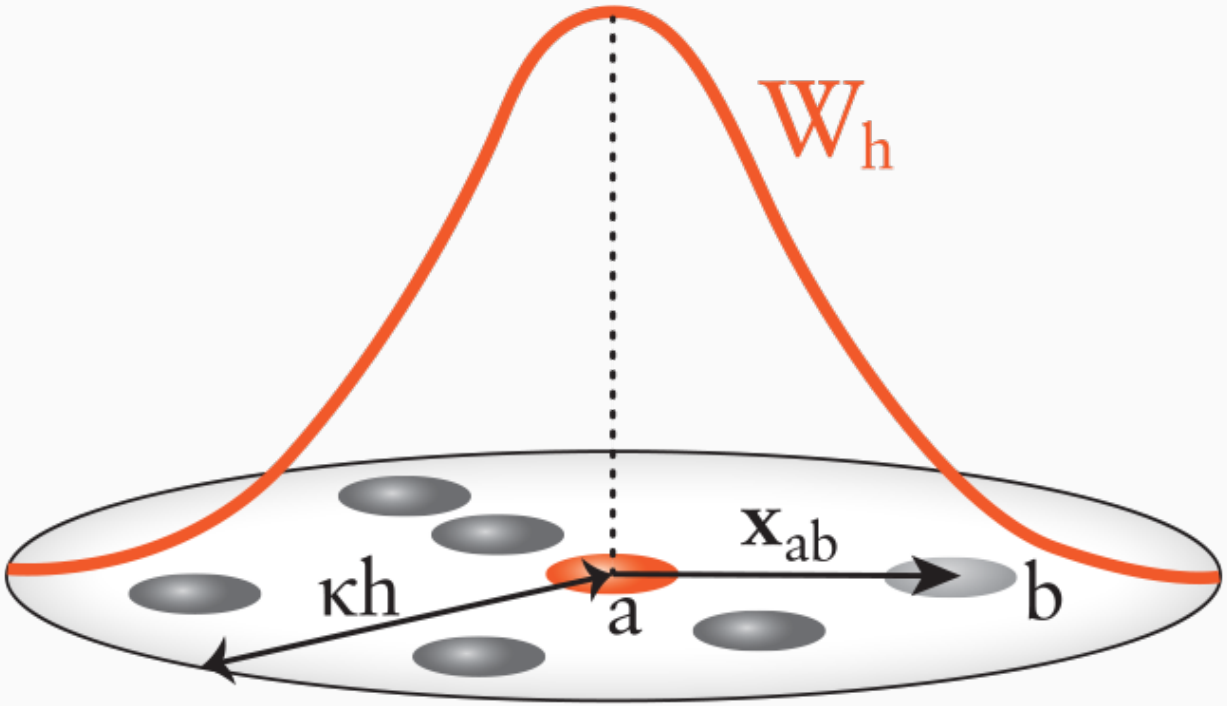


Figure 1: A schematic drawing of SPH kernel. κh is the size of kernel support. In case of Wendland's quintic kernel $\kappa = 2$.

About SPH - The underlying idea

SPH is based on the following idea:

Firstly we approximate δ - Dirac delta function using the smoothing kernel W_h. We can than write:

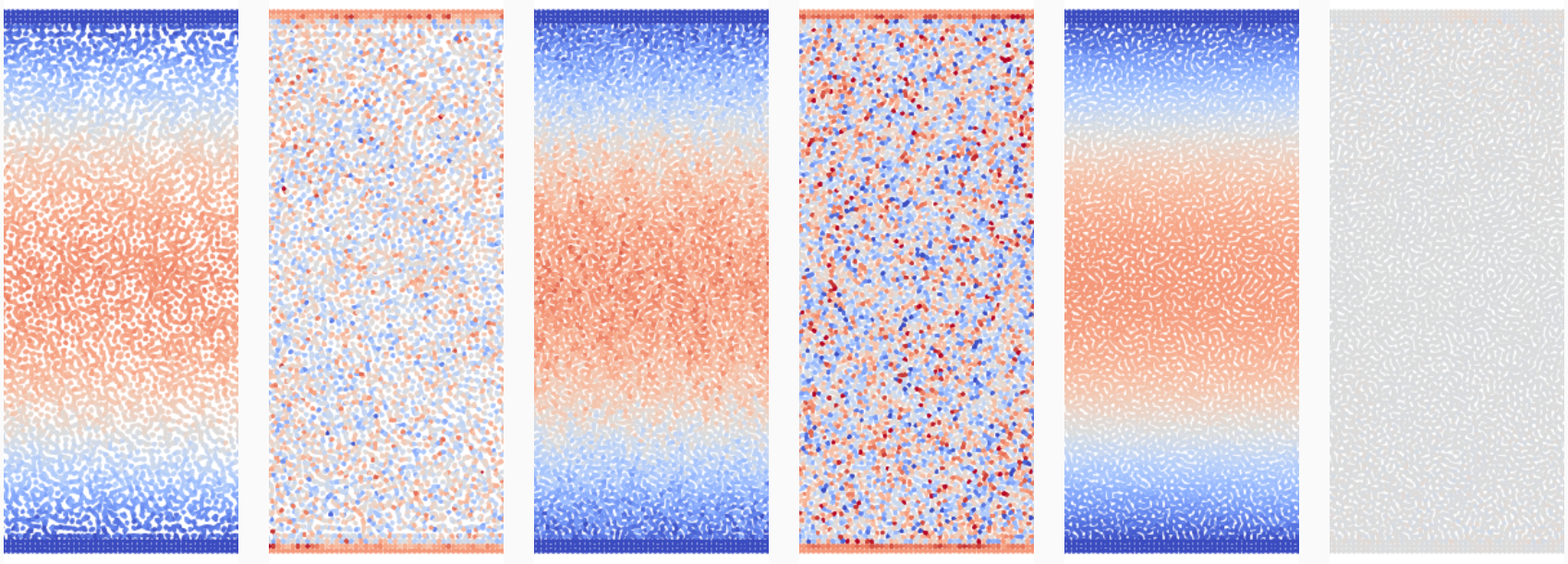
$$\begin{aligned} f &= f * \delta \approx f * W_{h,ab} \\ \partial_x f &= \partial_x f * \delta \approx \partial_x f * W_{h,ab} = f * \partial_x W_{h,ab}. \end{aligned} \tag{5}$$

Secondly, we approximate the integral in convolution using numeric quadrature:

$$\begin{aligned} f * W_h(\mathbf{x}_a) &= \int f(\mathbf{y}) W_h(\mathbf{x}_a - \mathbf{y}) d\mathbf{y} \approx \sum_b V_b f_b W_{h,ab}, \\ f * \partial_{x_i} W_h(\mathbf{x}_a) &= \int f(\mathbf{y}) \partial_{x_i} W_h(\mathbf{x}_a - \mathbf{y}) d\mathbf{y} \approx \sum_b V_b f_b \partial_{x_i} W_{h,ab} \end{aligned} \tag{6}$$

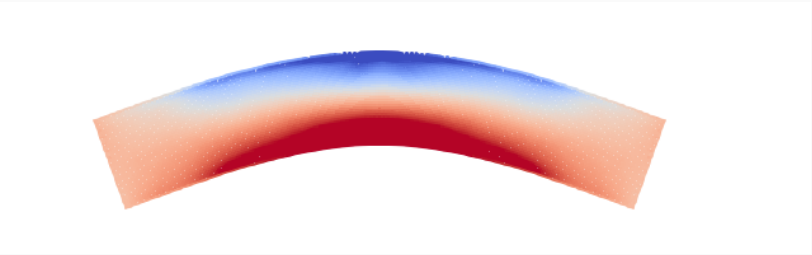
Results (so far)

Flow in a channel

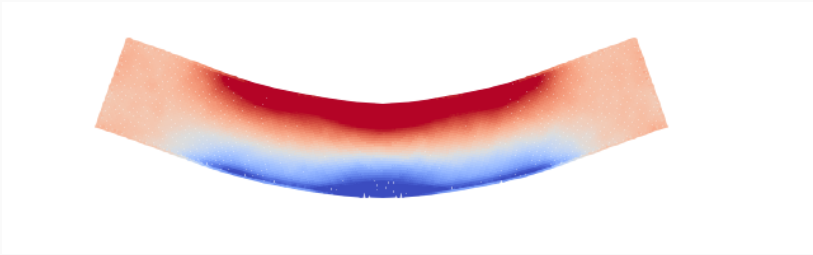


Comparison of three velocity (left) and pressure fields (right) with various numerical stabilizations.

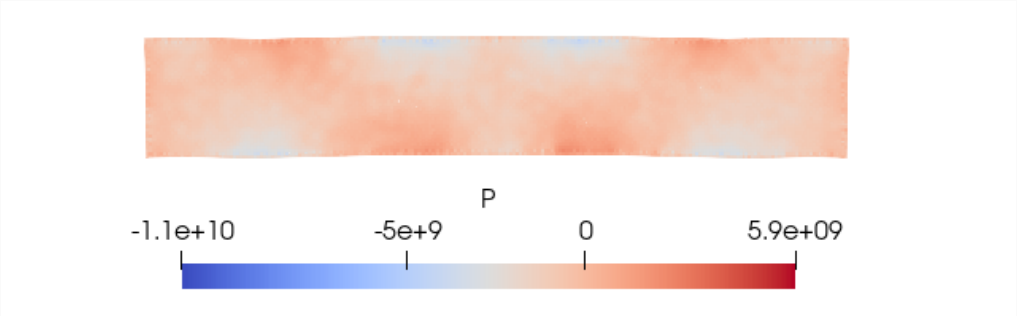
Oscillating beam



$t = 6.0 \times 10^{-6}$



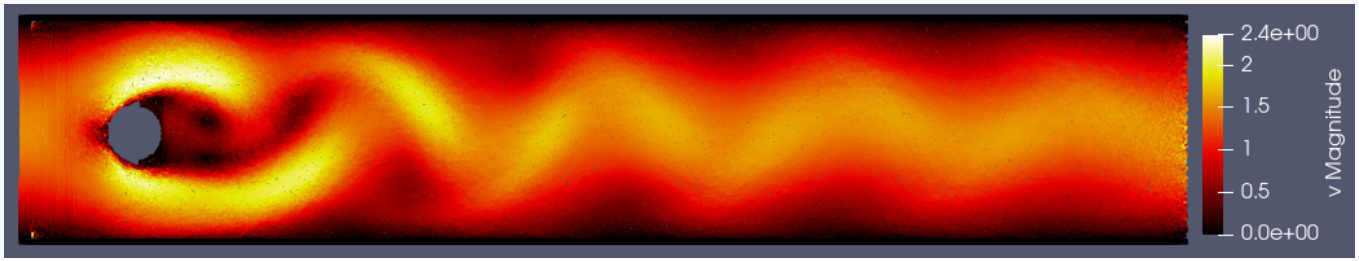
$t = 2.1 \times 10^{-5}$

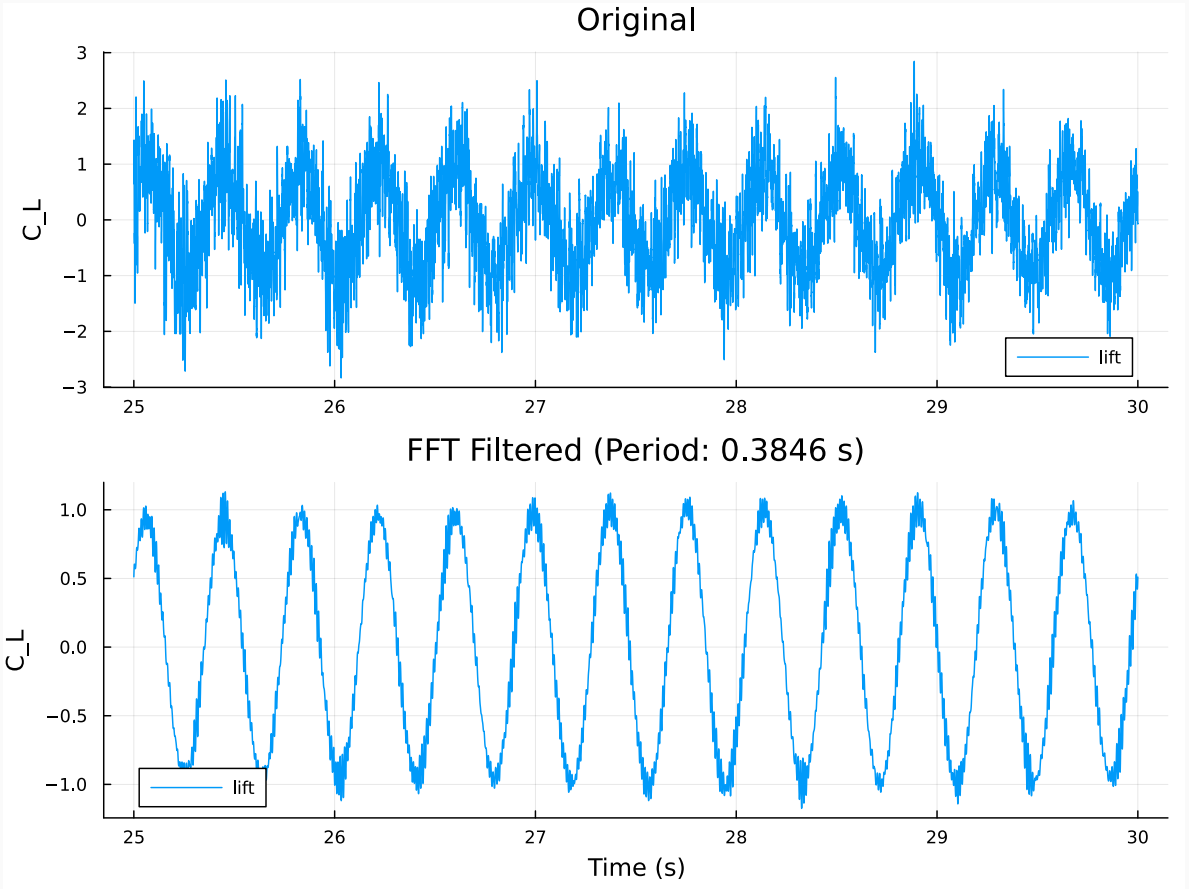


$t = 2.82 \times 10^{-5}$

The pressure field in the beryllium plate at three different times during the simulation.

Flow around cylinder





Tensile instability

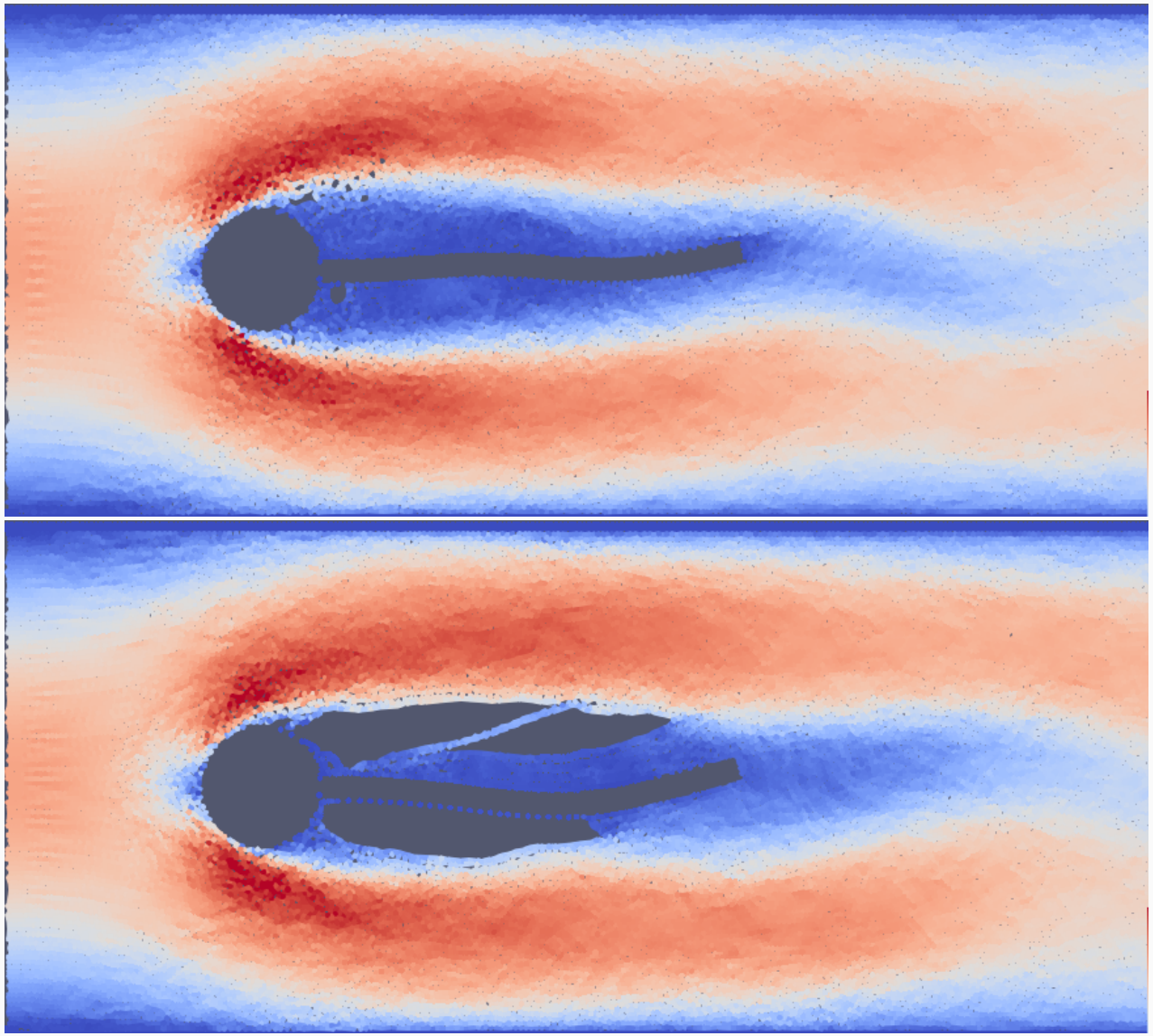
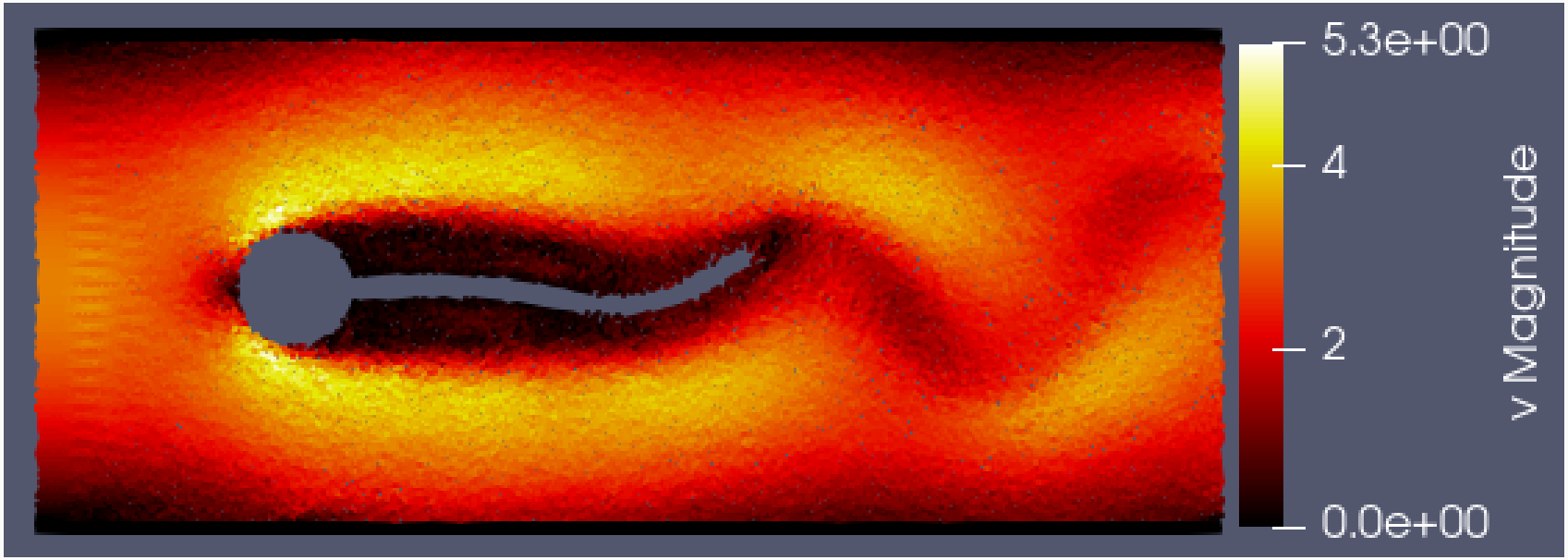
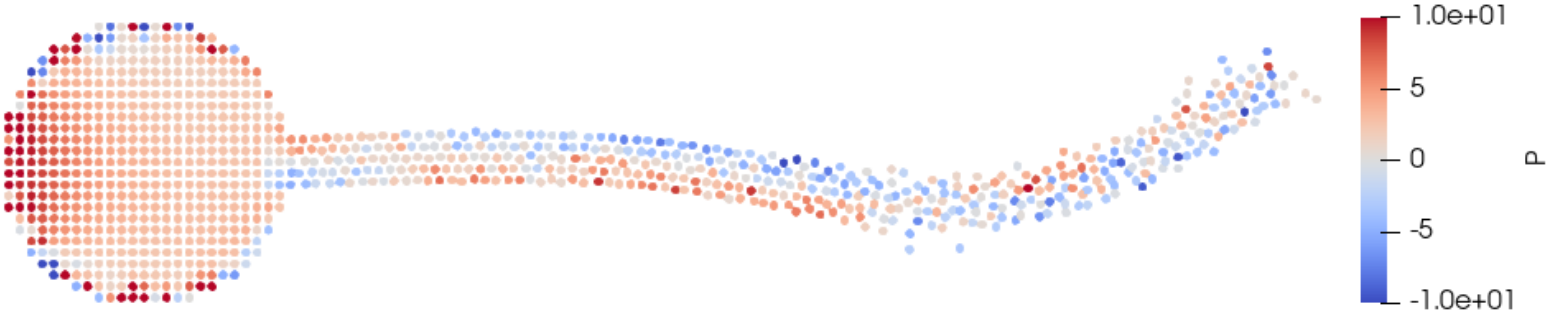


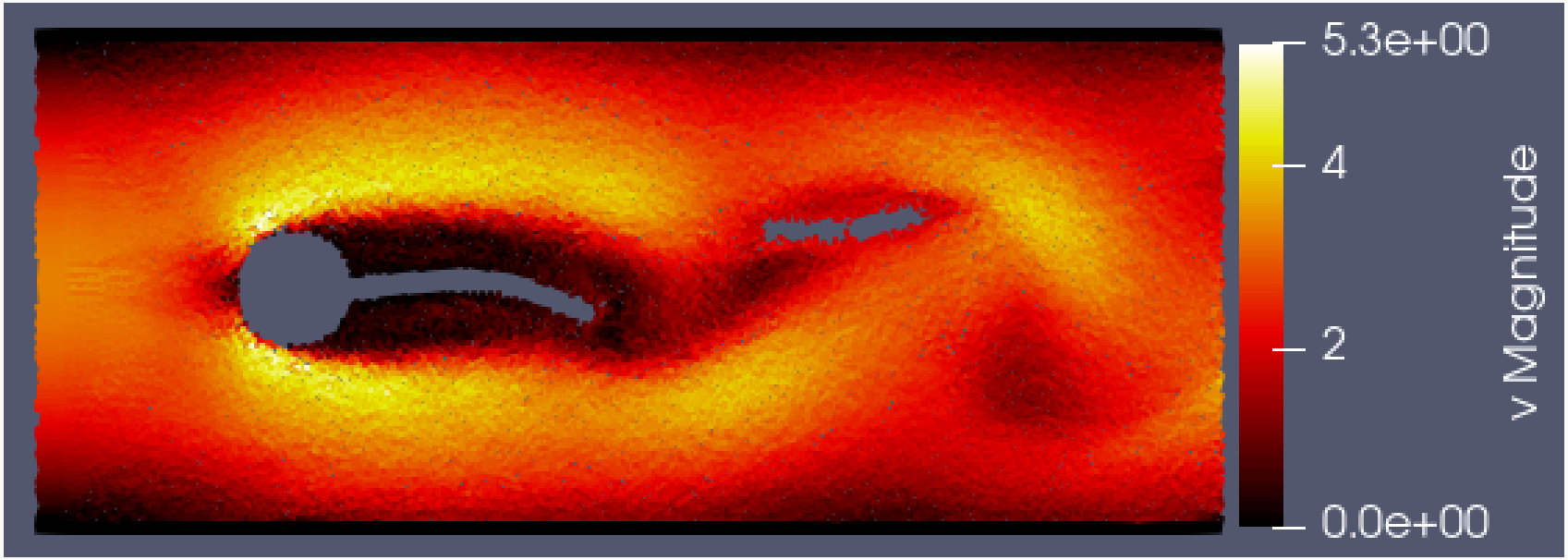
Figure 2: Tensile instability

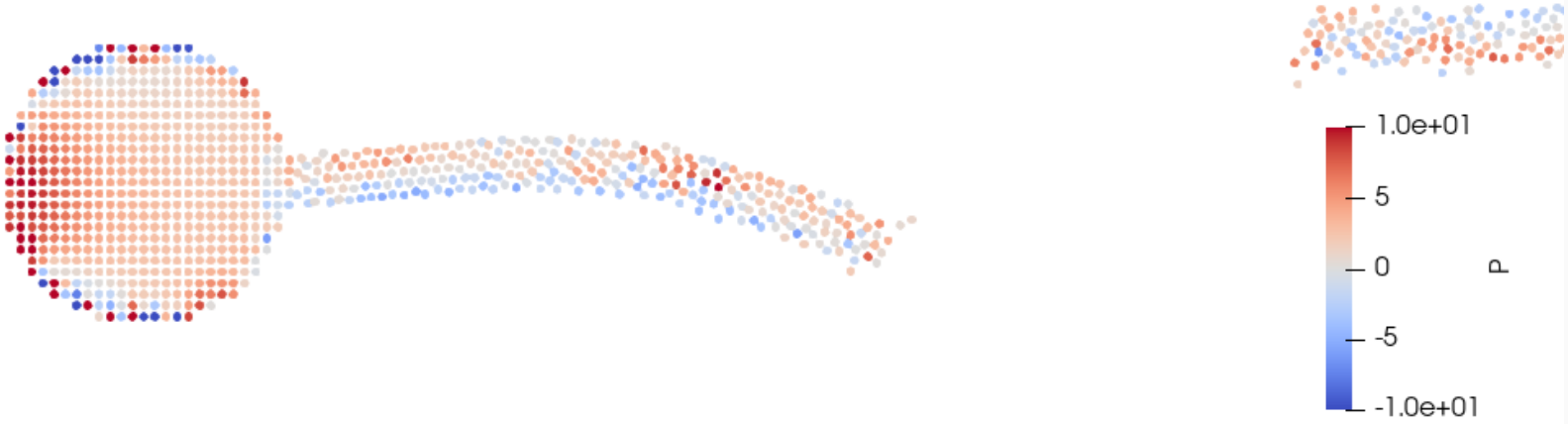
Turek-Hron benchmark





Turek-Hron benchmark





Questions

The distortion field A

$$\mathrm{d}\mathbf{A}_a = -\mathbf{A}_a \left(\sum_b m_b \mathrm{d}\mathbf{x}_{ab} \otimes \nabla W_{ab} \right) \left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla W_{ab} \right)^{-1} \tag{7}$$

There is no guarantee that the right-hand side is integrable. Therefore, in general we have for a closed loop Γ:

$$\int_\Gamma \mathrm{d}\mathbf{A}_a \neq 0 \tag{8}$$

This means some local residual stresses and artificial inelasticity.

What are the factors that influence the pressure "noise", apart from penalty terms?

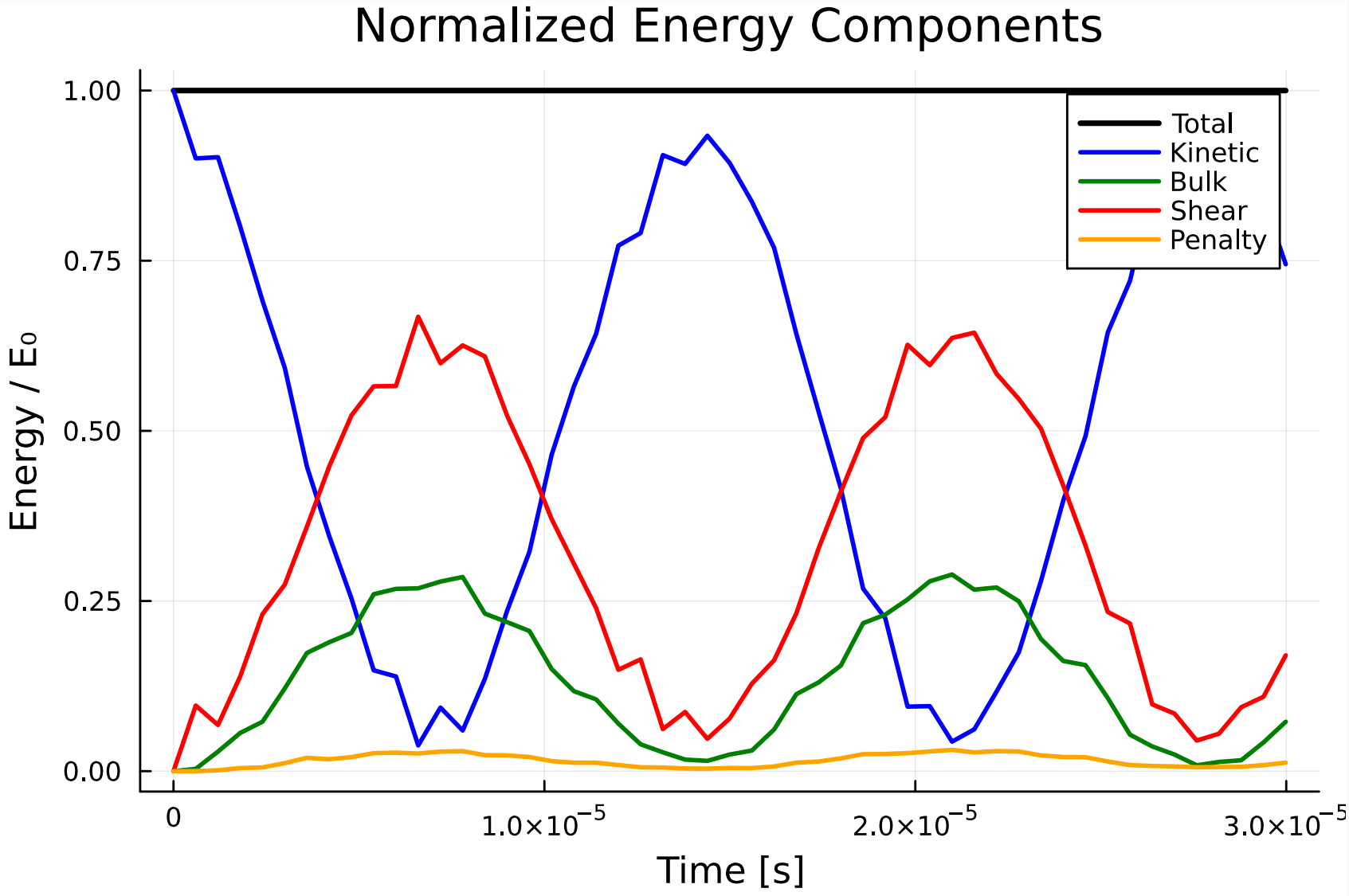
What factors play role in pressure "noise":

- Kernel & resolution
- numerical sound speed
- boundary treatment
- Reynolds number
- time explicit solver

How good are the conservational properties after all the corrections?

- Adding the tensile penalty only shifts the energy by a constant margin.
- Moreover it can be shown, that if carefully chosen, adding these terms to the system still preserves the conservational properties.

How good are the conservational properties after all the corrections?



Fluctuations of particle density are not just noise, but also occur naturally. Could it be possible to somehow align the observed fluctuations with the natural ones?

- In standard SPH methods the fluctuations are treated as a noise.
- Smoothed dissipative particle dynamics [**?**] adds fluctuations to SPH
- A new discretization of **A** from [**?**] can be used to add fluctuations to SPH:

$$D = \sum_{\alpha} \sum_{\beta} W_{\alpha} W_{\beta} ||\mathbf{X}_{\alpha\beta} - \mathbf{A} \cdot \mathbf{x}_{\alpha\beta}||^2 \tag{9}$$

- This formulation will be the aim of future student project [**?**]

Is it possible to e.g. model anisotropic and more complicated elastic energy functionals using this approach? Are there limitations?

SHTC can handle various problems - multiphase flows, inviscid and viscous Newtonian fluid, non-linear hyperelastic and elastoplastic solids, porous media etc.
Anisotropy is possible, if one specifies the state variables and equips each particle with *orientation vector* $\mathbf{n}$:

$$e(\mathbf{A}, \mathbf{n}) = e_{isotropic}(\mathbf{A}) + e_{anisotropic}(\mathbf{A}, \mathbf{n}) \tag{10}$$

$$\frac{D\mathbf{n}}{Dt} = \Omega \times \mathbf{n} \tag{11}$$

where $\Omega = \frac{1}{2}\nabla \times \mathbf{v}$.

Are there any good analytical error estimates?

It can be shown that for $W(\mathbf{x})$ such that supp $W(\mathbf{x}) \subset B_h(0)$ the following equality holds:

$$(f * W)(t, \mathbf{x}) = f(t, \mathbf{x}) + O(h^2) \tag{12}$$

Similarly

$$(f * \nabla W)(t, \mathbf{x}) = \nabla f(t, \mathbf{x}) + O(h^2) \tag{13a}$$

$$(\mathbf{F} * \nabla W)(t, \mathbf{x}) = \int_\Omega \mathbf{F}(t, \mathbf{x}') \cdot \nabla W(\mathbf{x} - \mathbf{x}') \mathrm{d}\mathbf{x}' = \nabla \cdot \mathbf{F}(t, \mathbf{x}) + O(h^2) \tag{13b}$$

where f and $\mathbf{F}$ are scalar and vector functions respectively.
Better estimates exist, but they assume uniform distribution of particles, etc.